

Sets and Subsets

Set - A collection of objects. The specific objects within the set are called the elements or members of the set. *Capital letters* are commonly used to name sets.

Examples: Set $A = \{a, b, c, d\}$ or Set $B = \{1, 2, 3, 4\}$

Set Notation - Braces $\{ \}$ can be used to list the members of a set, with each member separated by a comma. This is called the “Roster Method.” A description can also be used in the braces. This is called “Set-builder” notation.

Example: Set A: The natural numbers from 1 to 10.

Members of A: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Set Notation: $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Set Builder Not.: $\{x | x \text{ is a natural number from 1 to } 10\}$

Roster Method

Representation of a Set

Sets can be represented in two ways –

- Roster or Tabular Form
- Set Builder Notation

Roster or Tabular Form

- The set is represented by listing all the elements comprising it. The elements are enclosed within braces and separated by commas.

Example 1 – Set of vowels in English alphabet, $A = \{a, e, i, o, u\}$

Example 2 – Set of odd numbers less than 10, $B = \{1, 3, 5, 7, 9\}$

Set Builder Notation

- The set is defined by specifying a property that elements of the set have in common. The set is described as $A = \{x : p(x)\}$

Example 1 – The set $\{a, e, i, o, u\}$

is written as –

$A = \{x: x \text{ is a vowel in English alphabet}\}$

Example 2 – The set $\{1, 3, 5, 7, 9\}$

is written as –

$B = \{x: 1 \leq x < 10 \text{ and } (x \% 2) \neq 0\}$

If an element x is a member of any set S , it is denoted by $x \in S$

and if an element y is not a member of set S , it is denoted by $y \notin S$.

Example – If $S = \{1, 1.2, 1.7, 2\}$, $1 \in S$ but $1.5 \notin S$

Ellipsis - Three dots (...) used within the braces to indicate that the list continues in the established pattern. This is helpful notation to use for *long lists* or *infinite lists*. If the dots come at the end of the list, they indicate that the list goes on indefinitely (i.e. an infinite set).

Examples: Set A : Lowercase letters of the English alphabet

Set Notation: $\{a, b, c, \dots, z\}$

Cardinality of a Set – The number of *distinct* elements in a set.

Example: Set A : The days of the week

Members of Set A : Monday, Tuesday, Wednesday,
Thursday, Friday, Saturday, Sunday

Cardinality of Set $A = n(A) = 7$

Equal Sets – Two sets that contain exactly the same elements, regardless of the order listed or possible repetition of elements.

Example: $A = \{1, 1, 2, 3, 4\}$, $B = \{4, 3, 2, 1, 2, 3, 4, \}$.

Sets A and B are equal because they contain exactly the same elements (i.e. 1, 2, 3, & 4). This can be written as $A = B$.

Equivalent Sets – Two sets that contain the same number of distinct elements.

Example:

$A = \{\text{Football, Basketball, Baseball, Soccer}\}$

$B = \{\text{penny, nickel, dime, quarter}\}$

Both Sets have 4 elements

$$n(A) = 4 \quad \text{and} \quad n(B) = 4$$

A and B are Equivalent Sets, meaning $n(A) = n(B)$.

Note: If two sets are Equal, they are also Equivalent!

Example:

Set $A = \{a, b, c, d\}$

Set $B = \{d, d, c, c, b, b, a, a\}$

Are Sets A and B Equal?

Sets A and B have the exact same elements!
 $\{a, b, c, d\}$

☐ Yes!

Are Sets A and B Equivalent?

Sets A and B have the exact same number of distinct elements!
 $n(A) = n(B) = 4$

☐ Yes!

The Empty Set (or Null Set) – The set that contains **no elements**.

It can be represented by either $\{ \}$ or $\emptyset$.

Note: Writing the empty set as $\{\emptyset\}$ is **not correct**!

Symbols commonly used with Sets –

$\in \rightarrow$ indicates an object is an **element** of a set.

$\notin \rightarrow$ indicates an object is **not** an element of a set.

$\subset \rightarrow$ indicates a set is a **subset** of another set.

$\subset \rightarrow$ indicates a set is a **proper subset** of another set.

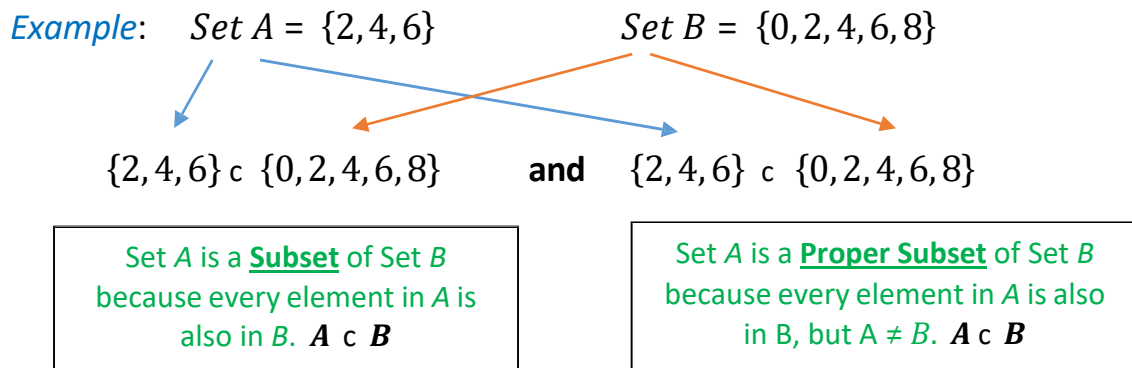
$\cap \rightarrow$ indicates the **intersection** of sets.

$\cup \rightarrow$ indicates the **union** of sets.

Subsets – a set whose elements are found in another set.

For Sets A and B , Set A is a **Subset** of Set B if every element in Set A is also in Set B . It is written as $A \subset B$.

Proper Subsets – For Sets A and B , Set A is a **Proper Subset** of Set B if every element in Set A is also in Set B , **but** Set A does not equal Set B . ($A \neq B$) It is written as $A \subset B$.



Note: The Empty Set is a Subset of every Set.

The Empty Set is also a Proper Subset of every Set except the Empty Set.

Number of Subsets – The number of distinct subsets of a set containing n elements is given by 2^n .

Number of Proper Subsets – The number of distinct proper subsets of a set containing n elements is given by $2^n - 1$.

Example: How many Subsets and Proper Subsets does Set A have?

Set $A = \{\text{bananas}, \text{oranges}, \text{strawberries}\}$

$n = 3$

Subsets = $2^n = 2^3 = 8$

Proper Subsets = $2^n - 1 = 7$

Example: List the **Proper Subsets** for the Example above.

1. $\{\text{bananas}\}$
2. $\{\text{oranges}\}$
3. $\{\text{strawberries}\}$
4. $\{\text{bananas}, \text{oranges}\}$
5. $\{\text{bananas}, \text{strawberries}\}$
6. $\{\text{oranges}, \text{strawberries}\}$
7. $\emptyset$

What Are Set Operations?

- Set operations describe the relationship between two or more sets. In math, a set is just a collection of objects.

These objects (more commonly referred to as elements) can take many forms, such as:

- Numbers
- Letters
- Variables
- Points
- Shapes
- Symbols

1. Union ($A \cup B$)

The union of two sets, A and B, is the set of distinct elements that are in Set A, Set B, or both A and B.

Example:

If $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6\}$, the union of Set A and Set B, $A \cup B = \{1, 2, 3, 4, 5, 6\}$.

2. Intersection ($A \cap B$)

The intersection of two sets, A and B, is the set of elements that are in BOTH Set A and Set B.

Example:

If $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 6, 9, 12, 15\}$, the intersection of Set A and Set B, $A \cap B = \{6\}$.

3. Difference ($A - B$)

The difference of two sets, $A - B$, is the set of elements that are unique to Set A. In other words, the difference includes the elements that are only in Set A and are not in Set B.

Example:

If $A = \{4, 8, 12, 16, 20\}$ and $B = \{4, 5, 16, 18, 20\}$, the difference of Set A and Set B, $A - B = \{8, 12\}$.

Intersection of Sets – The Intersection of Sets A and B is the set of elements that are in both A and B , i.e. what they have in common. It can be written as $A \cap B$.

Union of Sets – The Union of Sets A and B is the set of elements that are members of Set A , Set B , or both Sets. It can be written as $A \cup B$.

Example: Find the Intersection and the Union for the Sets A and B .

Set $A = \{Red, Blue, Green\}$

Set $B = \{Yellow, Orange, Red, Purple, Green\}$

Intersection: $A \cap B = \{Red, Green\}$

Union: $A \cup B = \{Red, Blue, Green, Yellow, Orange, Purple\}$

Set A and B only
have 2 elements in
common.

List each distinct element only once, even
if it appears in both Set A and Set B .

Try these on your own!

Given the set descriptions below, answer the following questions.

$U =$ All Integers from 1 to 10. $A =$ Odd Integers from 1 to 10,

$B =$ Even Integers from 1 to 10, $C =$ Multiples of 2 from 1 to 10.

1. Write each of the sets in roster notation.
2. What is the *cardinality* of Sets U and A ?
3. Are Set B and Set C Equal?
4. Are Set A and Set C Equivalent?
5. How many *Proper Subsets* of Set U are there?
6. Find B' and C'
7. Find $A \cup C'$
8. Find $B' \cap C$

Practice Set Operations

Exercise 1

Find the union between Set A and Set B , $A \cup B$.

$A = \{\text{blue, purple, orange, yellow, gray}\}$

$B = \{\text{blue, green, pink, black}\}$

Exercise 2

Find the intersection between Set A and Set B, $A \cap B$.

$$A = \{25, 30, 35, 40\}$$

$$B = \{23, 24, 25, 26, 27, 28, 29, 30\}$$

Exercise 3

Find the difference of Set A and B, $A - B$.

$$A = \{-1, 0, 3, 5\}$$

$$B = \{1, 2, 3, 4, 7, 12\}$$